

Can you prove the theorem formulated above?

Sol: **Proof:** Let a function $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined over the interval $[a, b]$. It's given that the function f is monotonic. Wlog, assume f is a nondecreasing function. $\therefore$ By defn, $\forall x_1, x_2 \in [a, b]$ where $a \leq x_1 < x_2 \leq b$, $f(x_1) \leq f(x_2)$.

$\because a \leq x_1, f(a) \leq f(x_1)$. $\therefore f(a)$ is the lower bound of f
 $\because x_2 \leq b, f(x_2) \leq f(b)$. $\therefore f(b)$ is the upper bound of f $\square$

Reader: Let the function $y = f(x)$ be defined over the closed interval $[a, b]$. We denote $f(a) = y_a$ and $f(b) = y_b$. To make the case more specific, let us assume that the function is nondecreasing. It means that $y_a \leq y_b$. I don't know how to proceed.

Author: Select an arbitrary point x over the interval $[a, b]$.

Reader: Since $a \leq x$ and $x \leq b$, then, according to the condition of the above theorem, $y_a \leq f(x)$ and $f(x) \leq y_b$. Thus, we get that $y_a \leq f(x) \leq y_b$ for all x in the domain of the function. This completes the proof.

Author: Correct. So, if a **monotonic** function is defined over a closed interval, it is bounded. As to a **nonmonotonic** function defined over a closed interval, it may be either bounded (Fig. 15a and f) or unbounded (Fig. 16b)

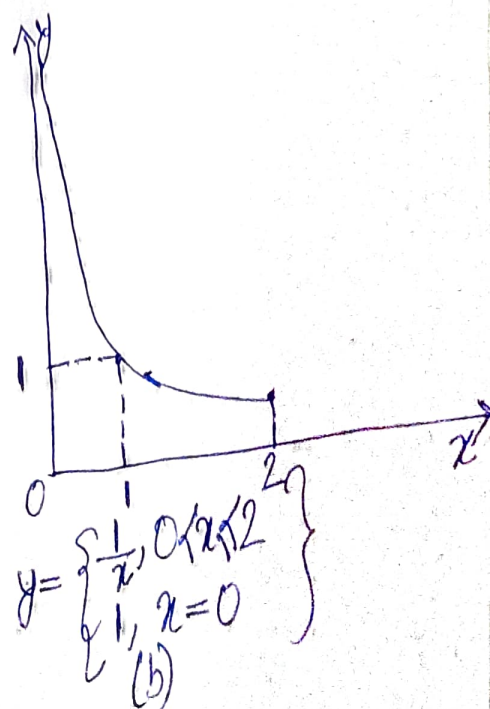
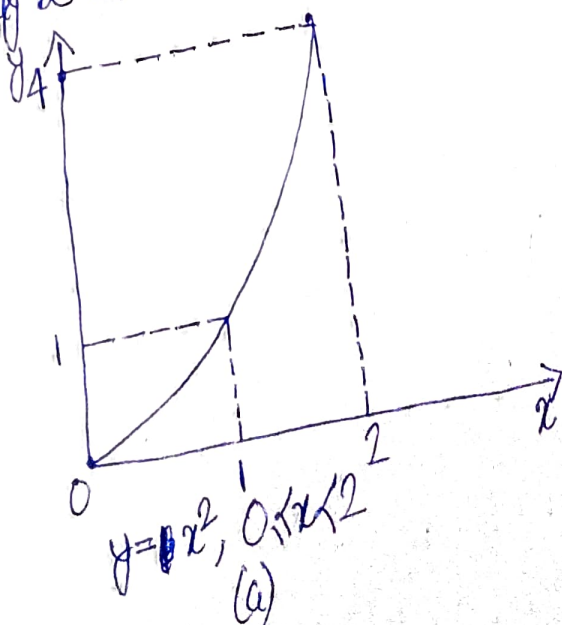


Fig. 16